Rooms With a View

Random Dorm Assignment and Cross-Class Friendship in College

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Fictional project created for the D-Lab AI Pulse demo, October 2026. Authors, university, data, and estimates are invented; cited papers are real.

Motivation: who you know predicts how far you go

- Chetty et al. (2022, *Nature*): 21 billion Facebook friendships, 72.2M users
- **Economic connectedness (EC)**: share of above-median-SES friends among below-median-SES people, ÷ 50%
- EC is among the **strongest predictors of upward mobility** ever measured
  - County correlation with upward mobility: **0.65**
  - Low-SES kids growing up with high-SES-level EC: incomes **+20%**
- Colleges are as income-segregated as neighborhoods (Chetty et al. 2020)
  - "Efforts to increase interaction across socioeconomic groups may be just as valuable at the college level"

The question

Does cross-class friendship **cause** better economic outcomes?

- Cross-sectional EC could proxy for everything else that differs across places
- Need variation in *exposure* to high-SES people that is unrelated to the person
- Colleges control one thing almost completely: **who lives next to whom**

This paper

Design

- Large public flagship ("the University")
- First-year rooms assigned by **lottery**
- Variation in high-SES share of floor-mates
- 4 cohorts, 24,871 students, 628 floors

Findings (per +10 pp high-SES floor-mates)

- EC **+0.07** (+10%)
- Paid internship **+2.1 pp**
- Earnings at 25 **+$2,350** (+6.1%)
- ∼70% runs through friendship
- Largest for first-gen, Black & Hispanic, low-EC-county students

Related literature

- **Social capital & mobility:** Chetty et al. (2022a,b) — EC, exposure vs. friending bias
- **Colleges & mobility:** Chetty et al. (2020) segregation across colleges; Chetty, Deming & Friedman (2023) Ivy-Plus vs. flagships; Zimmerman (2019); Michelman, Price & Zimmerman (2022)
- **Random roommates / dorms:** Sacerdote (2001); Marmaros & Sacerdote (2006); Carrell, Sacerdote & West (2013); Garlick (2018); Corno, La Ferrara & Burns (2022)
- **Mechanisms:** peers in own cohort matter (Chetty et al. 2024); race gaps (Chetty et al. 2020b)

Contribution: first causal estimate of the effect of EC on economic outcomes

Setting: the housing lottery

- First-years required to live on campus (unless local commuter)
- Housing Office assigns rooms by **computerized lottery**
  - Does not see income, race, major
  - No roommate requests for first-years
  - Floor changes in fall: $<1\%$
- Floor $\approx$ 40 students: shared lounge, bathrooms, RA, mandatory floor events
- Treatment: **share of floor-mates who are high-SES** (parental AGI in national top quintile)
- Sample: **7,412 Pell-eligible students**

Data

Source	Content	Coverage
Housing + admissions	Floor, room, SAT, HS GPA, race, first-gen	All 7,412
Financial aid (FAFSA)	Pell, parental AGI	All
ID-card swipes	Co-swipe network (Year 1, Year 3)	6,463 (Yr 1)
Senior exit survey	Friend nominations	3,038
Career services	Paid internships	All
State UI records	Earnings & employer at $\approx$25	3,412

- Entering cohorts 2014–2017; earnings in 7th year after entry (2021–2024)

Measuring economic connectedness on campus

- **Tie**: two students swipe into the same building within 2 minutes on $\geq$10 distinct days in a semester
  - Own residence-hall entrance swipes **excluded**
- Individual EC, following Chetty et al. (2022):

$$\text{EC}_i = \frac{\text{share of } i\text{'s ties who are high-SES}}{0.5}$$

- Mean Year-1 EC among Pell students: **0.68**
- Validated against senior-survey friend nominations

Randomization works

Characteristic	Coef. on high-SES share	s.e.
Female	0.011	(0.018)
First-generation	−0.014	(0.019)
Black / Hispanic	0.004 / −0.009	(0.011) / (0.016)
SAT (/100)	0.031	(0.044)
High-school GPA	0.006	(0.009)
Parental AGI ($000s)	−0.42	(0.61)
Home-county EC	0.004	(0.006)
Joint F-test p-value	0.47	

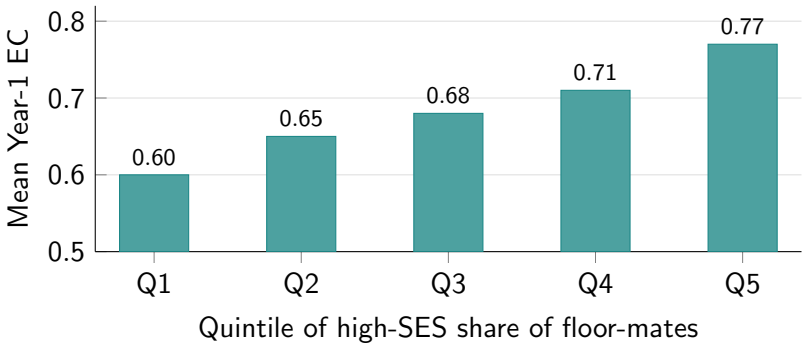
Peer composition is as good as random.

Empirical strategy

$$Y_{ifc} = \alpha + \beta\, \text{HighShare}_{f(-i)} + X_i'\gamma + \delta_c + \varepsilon_{ifc}$$

- $\text{HighShare}_{f(-i)}$: high-SES share of i's floor-mates (per 10 pp)
- δ_c: cohort fixed effects; X_i: predetermined controls (precision only)
- SEs clustered by floor (628 clusters)
- Mechanisms: 2SLS of Y on EC, instrumenting with HighShare

Exposure produces connection



Pell-eligible students, Year-1 network sample ($N = 6{,}463$). Quintile means of high-SES share: 0.31, 0.38, 0.43, 0.48, 0.55.

First stage: effect on economic connectedness

	EC Yr 1	EC Yr 1	EC Yr 3	EC survey
High-SES share (per 10 pp)	0.074***	0.071***	0.029**	0.031
	(0.011)	(0.010)	(0.012)	(0.020)
Controls	No	Yes	Yes	Yes
Mean	0.68	0.68	0.73	0.61
N	6,463	6,463	5,319	3,038

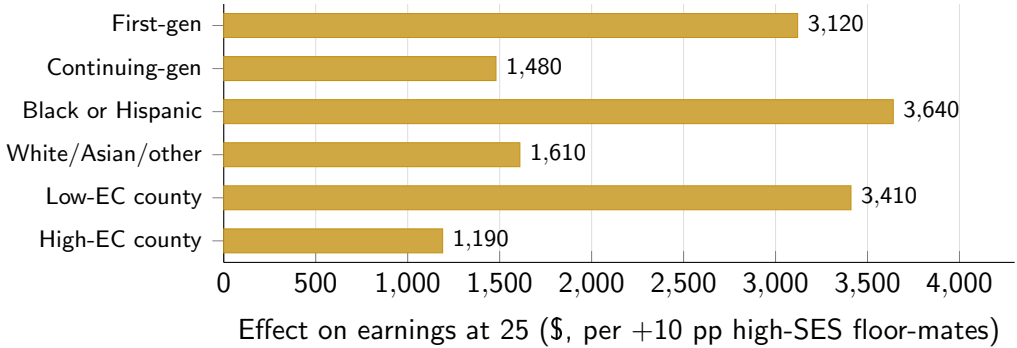
- +10 pp high-SES floor-mates ⇒ EC **+10%**
- Persists into Year 3; survey-based measure tells the same story

Early-career outcomes

Outcome	Mean	Effect (10 pp)	s.e.	N
Paid internship by junior year	0.37	0.021**	(0.009)	7,412
Internship at large employer	0.08	0.009*	(0.005)	7,412
Employed at high-wage firm, 25	0.21	0.019**	(0.009)	4,297
Earnings at 25 ($)	38,420	2,350**	(1,140)	3,412
Log earnings at 25	—	0.058**	(0.028)	3,412

- No effects on GPA, major, or graduation (see paper)
- Gains come from **where** students work, not *whether* they work

Who benefits most?



Effects concentrated among students who arrive with the **least social capital**

Robustness

- ✓ Lottery-block fixed effects
- ✓ Randomization inference ($p = 0.031$ for earnings)
- ✓ Wild cluster bootstrap
- ✓ Excluding living-learning communities
- ✓ Above-median SES threshold
- ✓ Count instead of share; floor-size controls
- ✓ Attrition not related to treatment

Headline results are stable across specifications.

Mechanism: it's the friendships

2SLS (effect of +0.1 EC)

- Internship: **+3.0 pp**
- High-wage firm: **+2.8 pp**
- Log earnings: **+0.085**
- First-stage $F \approx 40$

Mediation

- Controlling for EC shrinks earnings effect $0.058 \rightarrow 0.017$
- $\approx$ **70%** of effect via cross-class ties
- More internships found "through a friend" (+2.4 pp)

- No GPA or major effects $\Rightarrow$ not academic peer effects $\Rightarrow$ **friendship**

How big is this?

- 10th $\rightarrow$ 90th percentile floor (+23.6 pp high-SES): earnings **+14%**
- IV: ΔEC = 0.5 (10th–90th pct. county gap in Chetty et al.) $\Rightarrow$ earnings **+50%**
- Chetty et al. (2022) cross-section: same ΔEC $\approx$ +20% income
  - Our causal estimate is *larger* — consistent with attenuation in county-level EC
  - College = the moment career networks form

$\Rightarrow$ The EC–mobility relationship is **causal**

Takeaways and policy

- Random exposure to high-SES floor-mates raises low-SES students' EC, internships, and earnings (**+6.8%** at 25)
- Friendship is the channel
- **Policy:** SES-balanced first-year housing
  - Essentially zero cost
  - Scalable to ∼250 public universities with residential first-year programs
- **Next:** mid-career outcomes; less selective institutions; effects on high-SES students

Thank you!